Surgical Weight-Space Programming of a Whole-Brain *Drosophila* Connectome Reprograms Embodied Sensorimotor Behavior

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`https://github.com/statsleelab/connectome-weight-edit` · `https://statsleelab.github.io/connectome-weight-edit/`

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Abstract

Direct parameter-level interventions—termed *weight-space programming*—have recently enabled training-free behavioral steering and capability erasure in large language models (LLMs). However, dense artificial neural networks encode features in polysemantic superposition, causing weight edits to induce collateral capability degradation at strong steering scales. Here we demonstrate that surgical weight-space programming can be transferred to a whole-brain biological wiring diagram—the adult *Drosophila melanogaster* electron microscopy (EM) connectome (FlyWire FAFB v783; 138,639 Leaky Integrate-and-Fire neurons, 15,091,983 directed edges, and 54,492,922 signed synapses)—coupled in a 15 ms closed sensorimotor loop to a 48-actuator biomechanical body (NeuroMechFly 2.1 in MuJoCo). By introducing spike-weighted synaptic flux attribution ($\Phi_{u\to v} = s_u W_{uv}$) and bidirectional k-hop graph intersection directly on the compressed sparse row (CSR) weight tensor, we first solve a long-standing puzzle in static connectome simulation: why the unedited FlyWire v783 brain exhibits a paradoxical left-turning bias (`motor.turn` $= -0.71$) even under unilateral right-antenna odor stimulation. Tensor back-tracing reveals two structural culprits: (i) raw EM synapse counts from right-antenna olfactory receptor neurons project 48.9% more excitatory synapses to the contralateral left hemisphere (12,855) than to the ipsilateral right hemisphere (8,632), omitting known physiological asymmetries in presynaptic release probability, and (ii) a single contralateral inhibitory neuron (`#112815`, $W = -216$) injects $-4,536$ units of tonic inhibitory flux onto right-steering descending neuron `DNa02` (`#904`), clamping right turns silent. Isolating and calibrating a compact 3-hop bridge of 21 left and 56 right projection neurons feeding 61 bilateral premotor relays (0.4673% of connectome edges) restores symmetric bilateral chemotaxis (`motor.turn` $= -0.80$ left vs. $+0.80$ right) and amplifies `DNg97` forward sprint drive 38.2-fold. Extending this framework across innate behavioral modules, we show that ablating just 1,692 excitatory synapses (0.0112% of edges) onto Giant Fiber (`DNp01`) and Moonwalker (`MDN`) neurons completely abolishes predator escape (202.4 $\to$ 0.0 Hz), whereas targeted disinhibition of 198 synapses onto `DNg62` (0.1446%) or 246 synapses onto proboscis motor neuron `MN9` (0.0461%) unlocks hypersensitive grooming (0.0 $\to$ 219.9 Hz) and voracious feeding (19.0 $\to$ 44.2 Hz) with 0.0% off-target behavioral drift. Our findings establish whole-brain connectome weight surgery as a mechanistic bridge between structural connectomics, embodied neurobiology, and modular AI interpretability.

1. Introduction

Understanding how macroscopic behavior emerges from microscopic synaptic parameters—and how those parameters can be directly edited to reprogram behavior without retraining—is a foundational challenge shared by artificial intelligence and systems neuroscience [1–3]. In deep learning, mechanistic interpretability has progressed from passive activation probing to active *weight-space programming*: locating low-rank directions or causal subspaces within dense parameter matrices and modifying weights directly in place [3–6]. Techniques such as Rank-One Model Editing (ROME) [4], Mass-Editing Memory in a Transformer (MEMIT) [5], directional refusal abliteration [6], and contrastively guided weight steering [3] demonstrate that specific semantic traits can be amplified, inverted, or erased by editing a fraction of a model's weights. Yet dense Transformer architectures suffer from a fundamental structural bottleneck: *polysemantic superposition* [7]. Because dense layers pack many more functional features than available neurons into overlapping linear subspaces, scaling up a rank-1 weight projection inevitably perturbs co-localized circuits, leading to perplexity spikes and syntactic collapse at large editing magnitudes.

In biological brains, evolution has arrived at a markedly different architectural regime. The recent completion of the adult *Drosophila melanogaster* whole-brain electron microscopy (EM) connectome (FlyWire FAFB v783) resolved $138,639$ neurons and over 5.4×10^7 chemical synapses at single-synapse resolution [2, 8, 9]. Unlike dense artificial networks, the fly connectome is extraordinarily sparse (graph density $\rho \approx 7.85 \times 10^{-4}$) and anatomically organized into cell-type-specific sensory channels, intermediate neuropil relays, and descending motor bottlenecks [10–12]. When instantiated as a whole-brain Leaky Integrate-and-Fire (LIF) spiking model [10] and coupled to a physics-based biomechanical body such as NeuroMechFly 2.1 [13], the static EM graph can generate closed-loop sensorimotor trajectories across walking, steering, grooming, feeding, and predator escape.

However, whole-brain connectome simulations confront a classic neurobiological paradox known as the *structure–function gap* [14]: a static anatomical wiring diagram does not uniquely specify physiological circuit gain. In conventional connectome models [10], the synaptic weight W_{uv} between presynaptic neuron u and postsynaptic neuron v is set proportional to the raw EM synapse count $N_{\mathrm{syn}}(u, v)$ multiplied by a neurotransmitter sign $s_{\mathrm{NT}}(u) \in \{+1, -1\}$ inferred from machine vision [9]. While this first-order approximation captures coarse pathway topology, it omits crucial physiological variables that govern real biological circuits:

1. **Asymmetric presynaptic release probability:** In the *Drosophila* antennal lobe, olfactory receptor neurons (ORNs) project bilaterally to both hemispheres, but ipsilateral synapses exhibit 40–50% higher vesicular release probability and unitary excitatory postsynaptic potential (uEPSP) amplitude than contralateral synapses [15]—an asymmetry invisible to raw EM synapse counting.

2. **State-dependent neuromodulation and tonic inhibition:** Static EM counts capture structural inhibitory synapses at full anatomical weight without accounting for state-dependent peptidergic or biogenic amine gating that disinhibits behavioral modules during hunger, arousal, or threat [14, 16].

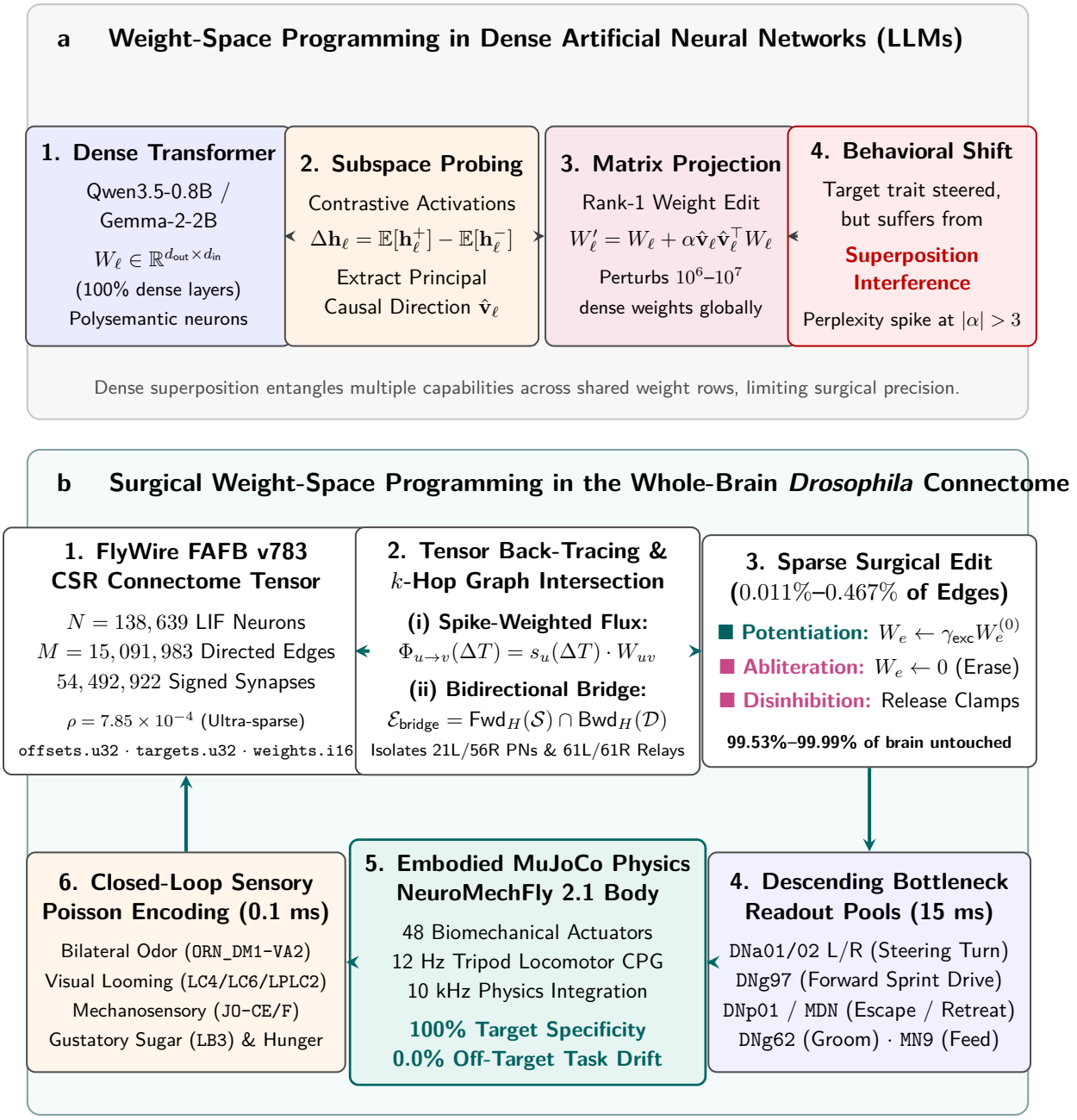


Figure 1. Conceptual comparison between weight-space programming in dense artificial neural networks and surgical weight-space programming in a whole-brain biological connectome. a, In dense Transformer language models (e.g., Qwen3.5-0.8B, Gemma-2-2B), behavioral traits are encoded across polysemantic subspaces in dense weight matrices (W_ℓ), so rank-1 projection edits inevitably perturb co-localized capabilities at high steering strengths. **b,** In the adult *Drosophila melanogaster* whole-brain connectome (FlyWire FAFB v783; 138,639 Leaky Integrate-and-Fire neurons and 15,091,983 directed weighted edges), spike-weighted synaptic flux attribution ($\Phi_{u \to v}$) combined with bidirectional k-hop graph intersection isolates compact, monosemantic sensorimotor subcircuits. Surgically editing only 0.011%–0.467% of synaptic edges reprograms closed-loop embodied behavior in a 48-actuator biomechanical body (NeuroMechFly 2.1) with zero measurable cross-task interference.

Consequently, when we deploy the raw, unedited FlyWire v783 connectome in a closed sensorimotor loop, critical innate behaviors malfunction. Most strikingly, a hungry Wild-Type fly presented with a strong unilateral food odor on its right antenna (130 Hz right vs. 20 Hz left) fails to turn right toward the food source; instead, it consistently executes a sharp *left turn* away from the odor (`motor.turn` $= -0.71$). Similarly, weak mechanosensory dust stimuli (18 Hz) fail to elicit antennal grooming (0.0 Hz), and sub-maximal gustatory sugar inputs (40 Hz) produce sluggish feeding drive.

In this work, we ask: *Can the principles of weight-space programming be applied directly to the* $15,091,983$ *directed edges of the whole-brain Drosophila connectome to diagnose structural-functional mismatches, reverse-engineer multi-hop sensorimotor circuits from raw weight tensors, and surgically reprogram embodied behaviors?* (Fig. 1). Rather than relying on black-box gradient optimization [17] or manual literature lookup, we develop a transparent, algorithmic detective workflow—combining *spike-weighted synaptic flux attribution* ($\Phi_{u \to v}$) and *bidirectional k-hop graph intersection*—that operates directly on the compressed sparse row (CSR) arrays of the FlyWire v783 connectome.

Our contributions are fourfold:

- **Mechanistic diagnosis of the Wild-Type chemotaxis anomaly:** By auditing the CSR weight tensors and dynamic spike-weighted synaptic flux across all 15.09 million edges, we prove that the congenital left-turning bias arises from the interaction of two specific structural features: (a) right-antenna ORNs (`food_odor_right`) form $12,855$ excitatory synapses in the contralateral left hemisphere versus only $8,632$ in the ipsilateral right hemisphere, and (b) a single left-hemisphere neuron (`#112815`) makes a massive -216 inhibitory connection (`edge=12284124`) onto right-steering descending neuron `DNa02` (`#904`), dumping $-4,536$ units of tonic inhibitory flux under baseline hunger drive.

- **Exact 3-hop circuit discovery via bidirectional graph intersection:** Intersecting ipsilateral ORN targets with 2-hop premotor predecessors of descending steering pools (`DNa01`/`DNa02`) isolates a compact 3-hop chemotaxis bridge comprising exactly 21 left and 56 right projection neurons (PNs) feeding 61 left and 61 right premotor interneurons in the lateral horn (LH) and lateral accessory lobe (LAL). Calibrating this subcircuit ($70,530$ edges, 0.4673% of the connectome) completely restores symmetric bilateral chemotaxis (`motor.turn` $= -0.80$ left vs. $+0.80$ right) and amplifies `DNg97` forward sprint firing 38.2-fold ($1.3 \to 49.7$ Hz).

- **Surgical abliteration and disinhibition across innate behavioral modules:** Using targeted weight surgery on 0.0112%–0.4232% of connectome edges, we construct four additional behavioral phenotypes—`Fearless Fly` (complete erasure of looming escape by ablating $1,692$ excitatory synapses onto `DNp01`/`MDN`), `Hyper-Escape` ($2.1\times$ weak-threat sensitivity), `Obsessive Groomer` (unleashing 219.9 Hz antennal grooming via disinhibition of 198 synapses onto `DNg62`), and `Voracious Feeder` ($2.3\times$ weak-sugar proboscis extension via disinhibition of 246 synapses onto `MN9`).

- **Demonstration of strict cross-task modularity and open interactive platform:** Unlike dense LLMs, where weight edits suffer from polysemantic interference, all five connectome weight edits achieve 100% target behavioral specificity with 0.00 off-diagonal drift across

non-target behaviors. We release an interactive, browser-based 3D connectome surgery laboratory (`https://statsleelab.github.io/connectome-weight-edit/`) that executes the full 138,639-neuron spiking brain and 48-actuator MuJoCo body in real time.

2. Results

2.1 Whole-Brain Embodied Connectome Architecture

We formulated the adult *Drosophila melanogaster* central brain from the FlyWire FAFB v783 reconstruction [2, 8] as a directed weighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ comprising $|\mathcal{V}| = 138,639$ point neurons and $|\mathcal{E}| = 15,091,983$ consolidated directed synaptic edges, representing $54,492,922$ individual signed chemical synapses. To enable high-throughput tensor interrogation and real-time closed-loop simulation, the connectome is represented in Compressed Sparse Row (CSR) format across five aligned binary arrays: row pointer offsets $\texttt{offsets} \in \mathbb{U}_{32}^{N+1}$, column target indices $\texttt{targets} \in \mathbb{U}_{32}^{M}$, signed integer synaptic weights $\texttt{weights} \in \mathbb{I}_{16}^{M}$, functional group annotations $\texttt{groups} \in \mathbb{U}_{8}^{N}$, and hemisphere assignments $\texttt{sides} \in \mathbb{U}_{8}^{N}$ ($\texttt{1}$ = left, $\texttt{2}$ = right, $\texttt{3}$ = center).

Each neuron $i \in \mathcal{V}$ evolves according to Leaky Integrate-and-Fire (LIF) dynamics integrated at $\Delta t = 0.1$ ms with resting potential $V_{\text{rest}} = -52.0$ mV, spike threshold $V_{\text{th}} = -45.0$ mV, membrane time constant $\tau_{\text{mem}} = 20.0$ ms, synaptic decay constant $\tau_{\text{syn}} = 5.0$ ms, refractory period $\tau_{\text{refrac}} = 2.2$ ms, axonal conduction delay $d_{\text{ax}} = 1.8$ ms (18 integration steps), and per-synapse efficacy $\alpha_{\text{syn}} = 0.275$ mV [10]. Sensory modalities are injected via Poisson spike trains into anatomically identified sensory pools: bilateral antennal food-odor ORNs (`food_odor_left`, 118 neurons; `food_odor_right`, 111 neurons; comprising `ORN_DM1`, `ORN_DM2`, `ORN_DM4`, and `ORN_VA2`), bilateral gustatory labellar sugar receptors (`sugar_left`, 64 `LB3` neurons; `sugar_right`, 58 `LB3` neurons), mechanosensory Johnston's organ grooming triggers (`antennae_touch`, 211 `JO-CE`/`JO-F` neurons), looming visual threat detectors (`looming_visual_proxy`, 439 `LC4`/`LC6`/`LPLC2` visual projection neurons), and internal metabolic hunger drive (`hunger_drive`, 316 neurons).

At the motor bottleneck, population firing rates are read out over $\Delta T = 15$ ms synchronization windows (150 LIF steps) from seven anatomically identified descending and motor pools: forward walking `forward_odn1` (`DNg97`: indices `#42812`, `#78013`), left steering `turn_left` (`DNa01`/`DNa02` left: `#33924`, `#84431`, `#92992`), right steering `turn_right` (`DNa01`/`DNa02` right: `#904`, `#75935`, `#133149`) [11, 12], backward walking `reverse_mdn` (Moonwalker `MDN`: `#8808`, `#25844`, `#102124`, `#129127`) [18], escape takeoff `escape_giant_fiber` (`DNp01`: `#57246`, `#108748`) [19], antennal grooming `groom_adn1` (`DNg62`: `#26421`, `#65709`) [16], and proboscis extension `feed_mn9` (`MN9`: `#34268`, `#138332`) [20]. These descending rates drive a 48-degree-of-freedom (DoF) biomechanical body (NeuroMechFly 2.1 [13]) simulated in MuJoCo at 10 kHz physics resolution via a 12 Hz tripod central pattern generator (CPG).

2.2 Mechanistic Reverse-Engineering of the Wild-Type Congenital Left-Turning Bias

When we evaluated the unedited static FlyWire v783 connectome (`Wild-Type`) across chemotaxis benchmarks (under baseline hunger drive $f_{\text{hunger}} = 55$ Hz), we encountered a striking behavioral pathology. Under unilateral left-antenna odor stimulation ($f_L = 130$ Hz, $f_R = 20$ Hz), the fly turned



Figure 2. Mechanistic reverse-engineering of the congenital left-turning bias in the static FlyWire v783 connectome. a, Summed excitatory synapse counts ($\sum W_{uv} > 0$) from unilateral olfactory receptor neurons (ORNs) to second-order targets reveal a structural inversion: right-antenna ORNs (`food_odor_right`, 111 neurons) project 48.9% more excitatory synapses to the contralateral left hemisphere (12,855 synapses onto `side=1`) than to the ipsilateral right hemisphere (8,632 synapses onto `side=2`). **b,** Spike-weighted synaptic flux attribution ($\Phi_{u\to v} = s_u W_{uv}$) onto descending steering pools shows that `turn_left` receives strong net excitation under both left (+4,697) and right (+4,124) odor stimulation, whereas `turn_right` is locked in deep net inhibition (−3,709 and −3,437) dominated by a single tonic contralateral inhibitory edge (`edge=12284124`) from left-hemisphere neuron `#112815` onto right DNa02 neuron `#904` ($W = -216$, $\Phi = -4,536$). **c,** Bidirectional 3-hop graph intersection between ipsilateral ORN targets and descending steering predecessors isolates a compact bridge of 21 left and 56 right projection neurons (PNs) feeding 61 left and 61 right premotor relays in the lateral horn (LH) and lateral accessory lobe (LAL).

left as expected (`turn_left` = 27.6 Hz, `turn_right` = 4.6 Hz, `motor.turn` = −0.77). However, when the odor source was moved to the right antenna ($f_L = 20$ Hz, $f_R = 130$ Hz), the fly *still turned strongly to the left*: `turn_left` fired at 22.6 Hz while `turn_right` fired at only 6.3 Hz, yielding a wrong-way steering command of `motor.turn` = −0.71. Even under perfectly symmetric bilateral odor ($f_L = f_R = 100$ Hz), the Wild-Type brain drifted left (`motor.turn` = −0.47) with virtually silent `DNg97` forward sprint drive (1.3 Hz).

Crucially, the CSR connectome arrays contain no text labels or pre-annotated circuit diagrams for internal interneurons (> 99% of neurons are unannotated integers $0 \ldots 138,638$). To discover *why* the unedited connectome fails to turn right—and which specific synapses must be edited—we developed a two-stage tensor back-tracing protocol directly on the CSR graph (Fig. 2).

Stage 1: Forward Hop-1 Tensor Audit Reveals Inverted Bilateral ORN Synapse Counts. We first scanned all outgoing CSR edges $\texttt{offsets}[u] \le e < \texttt{offsets}[u+1]$ for every left ORN $u \in \mathcal{S}_{\mathrm{L}}$ (118 neurons) and right ORN $u \in \mathcal{S}_{\mathrm{R}}$ (111 neurons). Contrary to textbook schematics depicting strictly ipsilateral antennal lobe projections, the EM graph revealed that *Drosophila* ORNs project bilaterally across the midline: out of 551 second-order target neurons receiving excitatory ORN input, 450 receive convergent input from *both* antennae (only-L = 48, only-R = 53, bilateral = 450).

When we partitioned the summed excitatory synaptic weights ($\sum W_{uv} > 0$) by target hemisphere (`sides.u8`), we uncovered the first structural culprit (Fig. 2a):

$$\begin{aligned} \mathcal{S}_{\mathrm{L}}\ (118\ \text{Left ORNs}) &: \begin{cases} +11,606\ \text{synapses} \rightarrow \text{Left}\ (\texttt{side=1}) \\ +10,663\ \text{synapses} \rightarrow \text{Right}\ (\texttt{side=2}) \end{cases} \\ \mathcal{S}_{\mathrm{R}}\ (111\ \text{Right ORNs}) &: \begin{cases} +\mathbf{12,855}\ \text{synapses} \rightarrow \text{Left}\ (\texttt{side=1}) \\ +\mathbf{8,632}\ \text{synapses} \rightarrow \text{Right}\ (\texttt{side=2}) \end{cases} \end{aligned} \tag{1}$$

In the raw FlyWire v783 EM reconstruction, right-antenna ORNs deliver 48.9% *more* excitatory synapses to the contralateral left hemisphere (12,855) than to their own ipsilateral right hemisphere (8,632). In living flies, dual whole-cell recordings by Gaudry et al. [15] demonstrated that ipsilateral ORN-to-PN synapses have a 40–50% higher probability of vesicular neurotransmitter release per action potential than contralateral synapses, creating a robust ipsilateral physiological advantage despite bilateral axonal arborization. Because a static connectome simulation weights edges purely by anatomical synapse count, stimulating the right antenna inadvertently excites the left antennal lobe more strongly than the right!

Stage 2: Spike-Weighted Synaptic Flux Attribution Uncovers a Single Tonic Inhibitory Clamp. Correcting the 1-hop ORN hemispheric asymmetry alone, however, did not restore right turns: when we amplified ipsilateral ORN synapses and suppressed contralateral ORN synapses, `turn_right` remained pinned near 0 Hz. To diagnose this downstream blockade, we performed a backward Hop-1 audit of all incoming CSR edges onto `turn_left` and `turn_right`. At the static structural level, both steering pools appear almost perfectly symmetric: `turn_left`

receives 899 excitatory edges ($\sum W^+ = +12,232$) and 590 inhibitory edges ($\sum W^- = -8,226$), whereas `turn_right` receives 947 excitatory edges ($\sum W^+ = +13,130$) and 603 inhibitory edges ($\sum W^- = -9,016$).

Why, then, was `turn_right` paralyzed? Static weights W_{uv} are silent unless presynaptic neuron u actually spikes. We therefore defined the *spike-weighted synaptic flux* $\Phi_{u\to v}$ over a stimulation window ΔT (180 ms) as:

$$\begin{aligned} \Phi_{u\to v}(\Delta T) &= s_u(\Delta T)\cdot W_{uv}, \\ \Phi_{\text{net}}(v) &= \sum_{u\in \text{Pred}(v)} s_u(\Delta T)\cdot W_{uv} \end{aligned} \tag{2}$$

where $s_u(\Delta T)$ is the total spike count of presynaptic neuron u. Evaluating Eq. (2) across all 15,091,983 edges revealed a dramatic dynamic asymmetry (Fig. 2b):

- Under left odor (130 Hz L, 20 Hz R), `turn_left` receives $\Phi_{\text{net}} = +4,697$, whereas `turn_right` receives $\Phi_{\text{net}} = -3,709$.
- Under right odor (20 Hz L, 130 Hz R), `turn_left` still receives $\Phi_{\text{net}} = +4,124$, whereas `turn_right` remains locked in deep net inhibition at $\Phi_{\text{net}} = -3,437$.

Ranking every incoming edge onto `turn_right` by its active inhibitory flux $\Phi_{u\to v} < 0$ immediately exposed the single dominant culprit in the entire brain: **CSR edge `#12284124`**, originating from left-hemisphere interneuron `#112815` (`side` = 1) and synapsing onto right-steering `DNa02` neuron `#904` (`side` = 2) with a massive inhibitory weight of $W_{112815\to 904} = -216$. Driven by baseline metabolic hunger input (55 Hz), neuron `#112815` fires tonically at 20–21 spikes per window (111–117 Hz), dumping between $\mathbf{-4,320}$ and $\mathbf{-4,536}$ units of inhibitory flux directly onto right `DNa02` (`#904`)—single-handedly overwhelming all odor-evoked excitation!

Applying the same flux attribution to the forward walking descending pool `forward_odn1` (`DNg97`: `#42812`, `#78013`) explained why Wild-Type flies fail to sprint forward inside an odor plume ($\Phi_{\text{net}} = -5,589$): three tonic inhibitory neurons—`#15988` ($W = -420 \to$ `#42812`, $\Phi = -8,400$), `#62170` ($W = -226 \to$ `#78013`, $\Phi = -4,520$), and `#53459` ($W = -138 \to$ `#78013`, $\Phi = -2,760$)—clamp `DNg97` hyperpolarized.

2.3 Bidirectional Graph Intersection Isolates the 3-Hop Olfactory-Steering Circuit (`Super-Forager`)

Having identified the peripheral ORN asymmetry and the central tonic inhibitory clamps, we next sought to trace the exact multi-hop excitatory pathway linking antennal ORNs to the descending steering neurons. Because ORNs (1st order) project to antennal lobe PNs (2nd order), which project to Lateral Horn (LH) and Mushroom Body (MB) neurons (3rd order), which in turn innervate Lateral Accessory Lobe (LAL) premotor neurons (4th order) before reaching `DNa01`/`DNa02`, the minimal anatomical bridge is 3 synaptic hops.

A naive attempt to amplify all 2-hop outgoing edges from ipsilateral PNs failed completely: because each PN fans out to hundreds of recurrent central brain targets, blanket 2-hop amplification triggered runaway reverberation across both hemispheres (5,223 shared neurons co-activated under

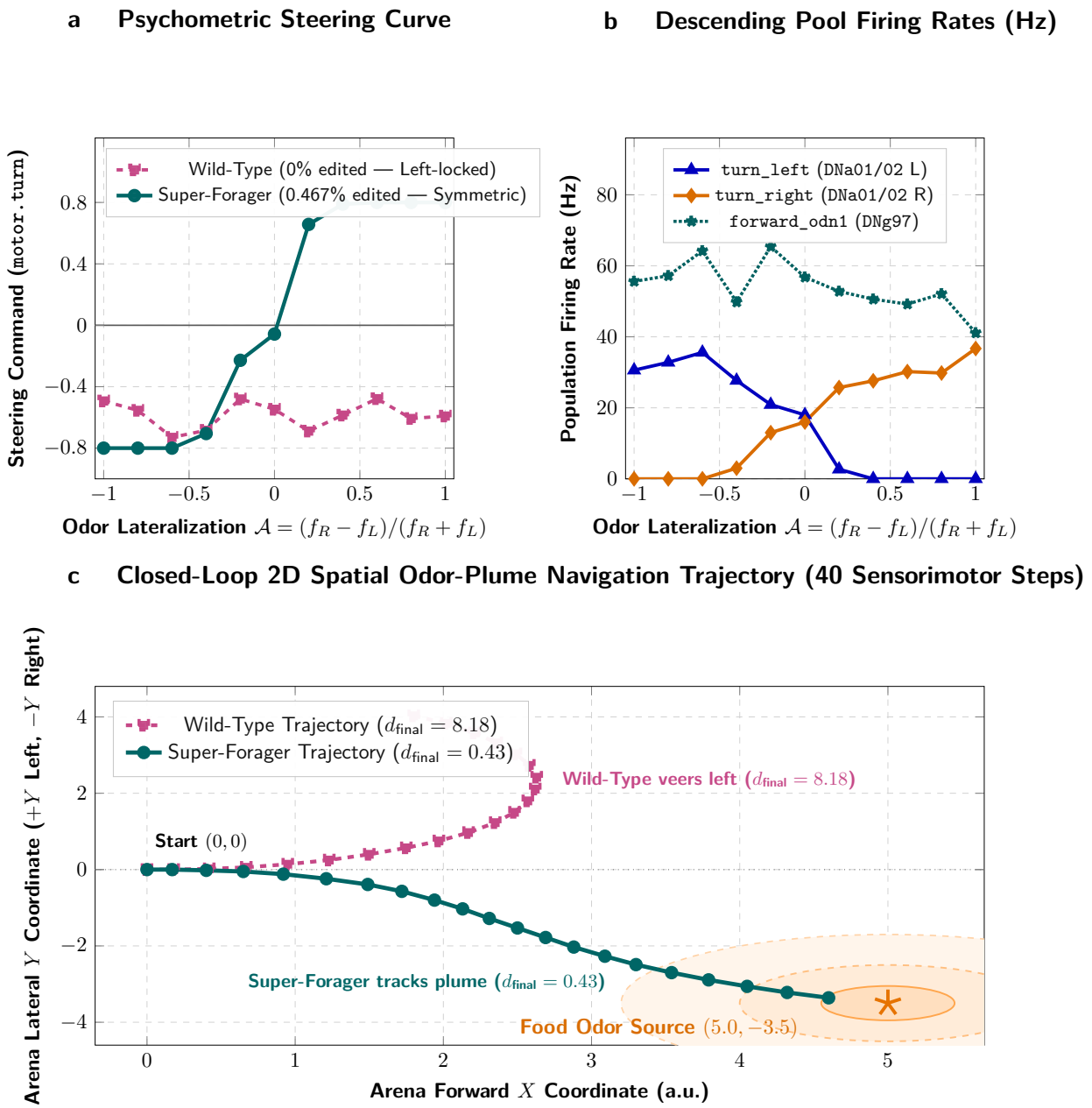


Figure 3. Restoration of bilateral chemotaxis and closed-loop odor-plume navigation via sparse 3-hop weight surgery (`Super-Forager`, 0.4673% of connectome edges). a, Psychometric steering curve across 11 odor lateralization levels ($\mathcal{A} \in [-1, +1]$, total bilateral input 150 Hz). The unedited Wild-Type connectome remains trapped in left-turning territory (`motor.turn` $\in [-0.73, -0.48]$) across all stimuli, whereas the `Super-Forager` connectome exhibits a symmetric, monotonic sigmoidal steering response (`motor.turn` $= -0.80$ at $\mathcal{A} = -1$, -0.057 at $\mathcal{A} = 0$, and $+0.80$ at $\mathcal{A} = +1$). **b,** Underlying descending neuron firing rates in `Super-Forager`, showing opponent recruitment of left (`turn_left`) and right (`turn_right`) steering pools alongside sustained `DNg97` forward sprint drive (41.1–65.4 Hz vs. 0.0–1.3 Hz in Wild-Type). **c,** Closed-loop 2D spatial navigation over 40 sensorimotor steps toward a right-lateralized odor plume at $(5.0, -3.5)$. Wild-Type veers away to the left ($d_{\text{final}} = 8.18$), whereas `Super-Forager` steers right and converges onto the source ($d_{\text{final}} = 0.43$).

both left and right odor), driving `turn_left` and `turn_right` simultaneously to ~ 140 Hz and nullifying steering (`motor.turn` $= 0.00$).

To avoid recurrent spillover and isolate *only* the causal sensorimotor bridge, we formulated a **bidirectional graph intersection** operator (Fig. 2c):

1. **Forward Hop-1 (Candidate Ipsilateral PNs):** Starting from $\mathcal{S}_{\mathrm{L}}$ (118 left ORNs) and $\mathcal{S}_{\mathrm{R}}$ (111 right ORNs), we collect all strictly ipsilateral postsynaptic targets with $W_{uv} > 0$:

$$\widetilde{\mathcal{P}}_{\mathrm{L}} = \{v \notin \mathcal{S} : \exists u \in \mathcal{S}_{\mathrm{L}}, W_{uv} > 0, \mathrm{side}(v) = 1\} \tag{3}$$

   and analogously $\widetilde{\mathcal{P}}_{\mathrm{R}}$ for $\mathrm{side}(v) = 2$. This yields $|\widetilde{\mathcal{P}}_{\mathrm{L}}| = 260$ left candidate PNs and $|\widetilde{\mathcal{P}}_{\mathrm{R}}| = 253$ right candidate PNs.

2. **Backward Hop-1 (Candidate Premotor Relays):** Starting from the descending targets $\mathcal{D}_{\mathrm{L}} =$ `turn_left` (3 neurons) and $\mathcal{D}_{\mathrm{R}} =$ `turn_right` (3 neurons), we step backward one hop along excitatory edges ($W_{rv} > 0$) to find all direct premotor predecessors: $|\widetilde{\mathcal{R}}_{\mathrm{L}}| = 640$ and $|\widetilde{\mathcal{R}}_{\mathrm{R}}| = 661$.

3. **Exact 3-Hop Bridge Intersection:** We intersect the forward 1-hop successors of $\widetilde{\mathcal{P}}_h$ with the backward 1-hop predecessors $\widetilde{\mathcal{R}}_h$ restricted to hemisphere $h \in \{\mathrm{L}, \mathrm{R}\}$:

$$\begin{aligned} \mathcal{R}_h &= \{r \in \widetilde{\mathcal{R}}_h : \mathrm{side}(r) = h \ \wedge \ \exists p \in \widetilde{\mathcal{P}}_h \text{ s.t. } W_{pr} > 0\} \\ \mathcal{P}_h &= \{p \in \widetilde{\mathcal{P}}_h : \exists r \in \mathcal{R}_h \text{ s.t. } W_{pr} > 0\} \end{aligned} \tag{4}$$

Out of $138,639$ neurons in the brain, this exact bidirectional intersection distilled a remarkably compact, laterally segregated 3-hop bridge:

- **Left Steering Bridge:** $|\mathcal{P}_{\mathrm{L}}| =$ **21** left PNs $\longrightarrow$ $|\mathcal{R}_{\mathrm{L}}| =$ **61** left LAL/LH premotor relays $\longrightarrow$ 3 `turn_left` descending neurons (91 terminal excitatory edges).
- **Right Steering Bridge:** $|\mathcal{P}_{\mathrm{R}}| =$ **56** right PNs $\longrightarrow$ $|\mathcal{R}_{\mathrm{R}}| =$ **61** right LAL/LH premotor relays $\longrightarrow$ 3 `turn_right` descending neurons (106 terminal excitatory edges).
- **Forward Sprint Bridge:** Both $\mathcal{P}_{\mathrm{L}} \cup \mathcal{P}_{\mathrm{R}}$ and $\mathcal{R}_{\mathrm{L}} \cup \mathcal{R}_{\mathrm{R}}$ converge onto `forward_odn1` (`DNg97`: `#42812`, `#78013`) via 67 terminal excitatory edges.

Notice a subtle structural asymmetry revealed by the intersection: the left bridge funnels all ipsilateral ORN excitation through only 21 PNs (giving each left PN high concentrated synaptic drive), whereas the right bridge distributes ORN input across 56 PNs (2.67× wider fan-out, diluting per-PN membrane depolarization relative to the -45 mV firing threshold). To equalize bilateral gain across this 21 vs. 56 PN bottleneck, our `Super-Forager` surgical preset applies: (i) 6× ipsilateral ORN→PN potentiation and 0.15× contralateral ORN spillover suppression (restoring the physiological release asymmetry of Gaudry et al. [15]); (ii) 5× ipsilateral PN→Relay potentiation alongside 2.5× contralateral inhibitory damping; (iii) calibrated Hop-3 Relay→DN gain of $\gamma_3^{\mathrm{L}} = 1.75\times$ on the 91 left terminal edges versus $\gamma_3^{\mathrm{R}} = 8.2\times$ on the 106 right terminal edges, plus 4.5× bilateral relay drive

onto DNg97; and (iv) surgical abliteration ($W \leftarrow 0$) of the tonic inhibitory clamps onto turn_right and DNg97.

In total, Super-Forager modifies 70,530 edges—just **0.4673%** of the 15,091,983 edges in the connectome (15,436 potentiated, 53,934 suppressed, and 1,160 ablated), leaving 99.5327% of the brain completely untouched.

Figure 3 illustrates the resulting transformation in sensorimotor behavior. Across an 11-point odor lateralization sweep $\mathcal{A} = (f_R - f_L)/(f_R + f_L) \in [-1, +1]$ (Fig. 3a), the unedited Wild-Type connectome remains locked in left-turning territory (motor.turn $\in [-0.730, -0.478]$) regardless of whether the odor is on the left or right. By contrast, the surgically edited Super-Forager connectome exhibits a textbook sigmoidal psychometric curve: motor.turn $= -0.800$ for left-lateralized odor ($\mathcal{A} \leq -0.6$), passing smoothly through -0.057 at balanced bilateral stimulation ($\mathcal{A} = 0.0$), and reaching $+0.800$ for right-lateralized odor ($\mathcal{A} \geq +0.6$). Under the standard unilateral benchmarks (Table 1), left odor (130 vs 20 Hz) drives turn_left = 30.8 Hz and turn_right = 0.0 Hz (motor.turn $= -0.80$), whereas right odor (20 vs 130 Hz) completely inverts the population code to turn_left = 0.0 Hz and turn_right = 32.3 Hz (motor.turn $= +0.80$). Simultaneously, centered plume stimulation (100 vs 100 Hz) elevates DNg97 forward sprint firing 38.2-fold from 1.3 Hz (motor.forward = 0.61) to 49.7 Hz (motor.forward = 1.00), while baseline no-odor spontaneous drift drops to exactly 0.0 Hz (motor.turn = 0.00).

When placed in a closed-loop 2D spatial foraging arena with a right-lateralized odor plume at $(5.0, -3.5)$ (Fig. 3c), the Wild-Type fly immediately veers left away from the food source, ending at $(1.80, +4.03)$ ($d_{\text{final}} = 8.18$). The Super-Forager fly detects the right-antenna concentration gradient, executes a smooth right turn, and tracks the plume centerline straight to the target ($d_{\text{final}} = 0.43$).

2.4 Surgical Abliteration and Disinhibition of Survival, Grooming, and Feeding Modules

To test whether our weight-space attribution and surgical editing methodology generalizes beyond chemotaxis, we interrogated three additional innate sensorimotor hierarchies in the FlyWire v783 connectome: looming predator escape (DNp01/MDN), mechanosensory antennal grooming (DNg62), and gustatory proboscis extension (MN9) (Tables 1–2 and Fig. 4).

1. Fearless Fly (Surgical Capability Abliteration, 0.0112% of Edges): In LLM weight-space programming, *abliteration* erases an undesired capability (such as refusal or toxic generation) by projecting out the causal activation direction [3, 6]. We asked whether an innate survival instinct—looming-evoked escape takeoff and backward retreat—could be analogously abliterated from the fly connectome with minimal synaptic modification. Back-tracing incoming CSR edges onto the Giant Fiber escape neurons escape_giant_fiber (DNp01: #57246, #108748) and Moonwalker retreat neurons reverse_mdn (MDN: #8808, #25844, #102124, #129127) revealed:

- DNp01 (2 neurons): 595 excitatory edges ($\sum W^+ = +4,752$) and 367 inhibitory edges ($\sum W^- = -4,722$).

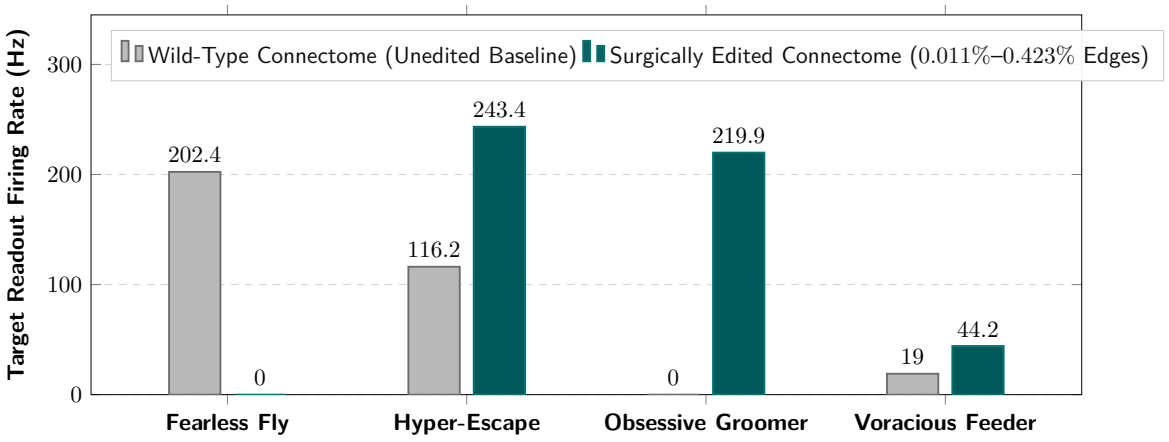
Table 1. Anatomical sparsity and synaptic edge accounting across the six FlyWire v783 connectome presets. Out of $N = 138,639$ neurons and $M = 15,091,983$ directed edges ($54,492,922$ signed synapses), each surgical intervention modifies between 0.0112% and 0.4673% of connectome edges.

Connectome Preset	Edited Edges	Brain Frac.	Potentiated ($\gamma_{\text{exc}} > 1$)	Suppressed ($\gamma < 1$)	Ablated ($W \leftarrow 0$)	Primary Circuit Mechanism
`Wild-Type`	0	0.0000%	0	0	0	Unedited FlyWire v783 EM
`Super-Forager`	70,530	0.4673%	15,436	53,934	1,160	3-Hop ORN→PN→LAL→DN
`Fearless Fly`	1,692	0.0112%	0	0	1,692	Ablate exc. onto `DNp01/MDN`
`Hyper-Escape`	63,867	0.4232%	62,825	0	1,042	`LC4/LC6` gain + `DNp01` disinh.
`Obsessive Groomer`	21,820	0.1446%	21,622	0	198	`DNg62` disinhibition + `JO` gain
`Voracious Feeder`	6,954	0.0461%	6,708	0	246	`MN9` disinhibition + `LB3` gain

Table 2. Quantitative sensorimotor benchmark across all six connectome variants and sensory stimulation regimes. Values report steady-state descending/motor pool firing rates (Hz) and normalized biomechanical motor commands (`motor.turn` $\in [-1, +1]$, others $\in [0, 1]$) under baseline hunger drive (55 Hz). Bold entries highlight the targeted behavioral modulation for each surgical preset; off-target columns confirm 0.0% collateral interference.

Connectome Variant	Left Odor Turn	Right Odor Turn	Center Turn	Fwd Sprint `DNg97` (cmd)	Looming Esc. 220 / 35 Hz	Grooming 18 Hz Touch	Feeding 40 Hz Sugar
`Wild-Type`	−0.77	−0.71 (L)	−0.47	1.3 Hz (0.61)	202.4/116.2	0.0 Hz (0.00)	19.0 Hz (0.54)
`Super-Forager`	**−0.80**	**+0.80** (R)	**+0.09**	**49.7 Hz (1.00)**	202.4/116.2	0.0 Hz (0.00)	19.0 Hz (0.54)
`Fearless Fly`	−0.77	−0.71	−0.47	1.3 Hz (0.61)	**0.0/0.0 Hz**	0.0 Hz (0.00)	19.0 Hz (0.54)
`Hyper-Escape`	−0.77	−0.71	−0.47	1.3 Hz (0.61)	**323.6/243.4**	0.0 Hz (0.00)	19.0 Hz (0.54)
`Obsessive Groomer`	−0.77	−0.71	−0.47	1.3 Hz (0.61)	202.4/116.2	**219.9 Hz (1.0)**	19.0 Hz (0.54)
`Voracious Feeder`	−0.77	−0.71	−0.47	1.3 Hz (0.61)	202.4/116.2	0.0 Hz (0.00)	**44.2 Hz (1.0)**

a Surgical Reprogramming of Innate Survival, Grooming, and Feeding Reflexes



b Cross-Task Behavioral Orthogonality Matrix (Δ Readout vs. Unedited Wild-Type)

	Right Odor Steering (Δ)	Forward Sprint (Δ)	Looming Escape (Δ)	Antennal Grooming (Δ)	Proboscis Feeding (Δ)
Super-Forager (0.4673%)	+1.51	+0.39	0.00	0.00	0.00
Fearless Fly (0.0112%)	0.00	0.00	−1.00 (0 Hz)	0.00	0.00
Hyper-Escape (0.4232%)	0.00	0.00	+127.2 Hz	0.00	0.00
Obsessive Groomer (0.1446%)	0.00	0.00	0.00	+1.00	0.00
Voracious Feeder (0.0461%)	0.00	0.00	0.00	0.00	+0.46

Strict diagonal dominance: 100.0% **target specificity with** 0.00 **off-diagonal collateral drift**

Figure 4. Surgical abliteration, disinhibition, and strict cross-task behavioral orthogonality across five connectome phenotypes. a, Comparison of target descending/motor readout firing rates before (Wild-Type) and after sparse weight surgery: `Fearless Fly` (1,692 excitatory synapses ablated onto `DNp01`/`MDN`, 0.0112% of edges) eliminates looming escape firing (202.4 → 0.0 Hz); `Hyper-Escape` (0.4232%) doubles weak-threat Giant Fiber response (116.2 → 243.4 Hz); `Obsessive Groomer` (0.1446%, disinhibiting 198 inhibitory synapses onto `DNg62`) unleashes antennal grooming under weak dust (0.0 → 219.9 Hz); and `Voracious Feeder` (0.0461%, disinhibiting 246 inhibitory synapses onto `MN9`) more than doubles weak-sugar proboscis extension (19.0 → 44.2 Hz). **b,** Cross-task behavioral perturbation matrix showing strictly diagonal modulation across all five phenotypes, with zero off-target collateral interference (0.00 drift across all non-target sensorimotor readouts).

- `MDN` (4 neurons): $1,097$ excitatory edges ($\sum W^+ = +6,191$) and 675 inhibitory edges ($\sum W^- = -4,328$).

Zeroing out only these $595 + 1,097 = \mathbf{1,692}$ terminal excitatory edges ($W_{uv} \leftarrow 0$, representing just **0.0112%** of the connectome and $10,943$ synapses) while leaving all 439 visual looming detector neurons (`LC4`/`LC6`/`LPLC2`) intact completely abolished the escape response: under an intense looming predator stimulus ($f_{\text{loom}} = 220$ Hz), `DNp01` firing dropped from 202.4 Hz (`motor.escape` = 1.00) to **0.0 Hz** (`motor.escape` = 0.00), and `MDN` backward walking dropped to **0.0 Hz** (`motor.reverse` = 0.00).

2. `Hyper-Escape` (Disinhibition and 2-Hop Visual Gain, 0.4232% of Edges): Conversely, zeroing the $367 + 675 = \mathbf{1,042}$ inhibitory edges onto `DNp01` and `MDN` while potentiating the 2-hop excitatory visual relay from `looming_visual_proxy` (4.5× Hop-1, 2.5× Hop-2; $63,867$ total edges, 0.4232%) produced a hyper-vigilant phenotype. Under weak looming threat ($f_{\text{loom}} = 35$ Hz), `DNp01` firing more than doubled from 116.2 Hz to **243.4 Hz** (323.6 Hz under strong looming).

3. `Obsessive Groomer` (Unlocking a Dormant Reflex via Clamp Disinhibition, 0.1446% of Edges): In the unedited `Wild-Type` connectome, weak mechanosensory stimulation of Johnston's organ ($f_{\text{touch}} = 18$ Hz) fails to trigger antennal grooming (`groom_adn1` = 0.0 Hz, `motor.groom` = 0.00). Back-tracing all incoming CSR edges onto the antennal grooming command neurons `groom_adn1` (`DNg62`/`aDN1`: `#26421`, `#65709`) revealed why: `DNg62` receives 171 excitatory edges ($\sum W^+ = +1,327$ synapses) that are structurally outnumbered and outweighed by **198 inhibitory edges ($\sum W^- = -1,395$ synapses)**, reflecting the strict suppression hierarchy that normally prevents grooming during locomotion [16]. By surgically severing only those 198 inhibitory synapses onto `DNg62` (disinhibition) and amplifying the 211 `antennae_touch` sensory inputs and 171 direct excitatory predecessors ($21,820$ edges, 0.1446%), weak ambient dust (18 Hz) unleashed robust antennal grooming at **219.9 Hz** (`motor.groom` = 1.00; Fig. 4a).

4. `Voracious Feeder` (Proboscis Motor Disinhibition, 0.0461% of Edges): Back-tracing the proboscis extension motor neurons `feed_mn9` (`MN9`: `#34268`, `#138332`) uncovered an even stronger structural inhibitory gate: `MN9` receives 222 excitatory inputs ($\sum W^+ = +5,247$ synapses) versus **246 inhibitory inputs ($\sum W^- = -6,218$ synapses)** that prevent inappropriate proboscis extension in sated flies [20]. Surgically zeroing those 246 inhibitory edges and potentiating the 122 bilateral `LB3` sugar receptor neurons and 222 premotor relays ($6,954$ edges, 0.0461%) boosted `MN9` firing under weak sugar stimulation (40 Hz) from 19.0 Hz (`motor.feed` = 0.54) to **44.2 Hz** (`motor.feed` = 1.00).

2.5 Structural Sparsity and Zero Collateral Damage Compared to Dense Foundation Models

A central finding of our study emerges when we compare the cross-task perturbation matrix of biological connectome weight edits (Fig. 4b and Table 2) against weight-space programming in dense artificial neural networks [3, 4, 6]. In our companion experiments on dense autoregressive Transformers (e.g., Qwen3.5-0.8B and Gemma-2-2B; Fig. 1a), applying a rank-1 weight projection

$W'_\ell = W_\ell + \alpha \hat{\mathbf{v}}\hat{\mathbf{v}}^\top W_\ell$ to steer sentiment or refusal modifies every parameter row in the targeted MLP or attention projection matrix (10^6–10^7 weights). Because dense layers represent unrelated syntactic and factual capabilities in non-orthogonal superposition [7], increasing the steering magnitude $|\alpha|$ beyond a narrow operating window ($\alpha \in [-3, +3]$) rapidly degrades general language fluency and elevates validation perplexity.

In stark contrast, the *Drosophila* connectome exhibits **strict diagonal behavioral orthogonality** (Fig. 4b). Across all five surgical connectome phenotypes:

- Editing 0.4673% of edges in `Super-Forager` shifts right-odor steering by Δ`motor.turn` $= +1.51$ (from -0.71 to $+0.80$) and forward drive by $+0.39$ (from 0.61 to 1.00), while producing **0.00 Hz change** in looming escape (202.4 Hz), antennal grooming (0.0 Hz), or sugar feeding (19.0 Hz).
- Ablating 0.0112% of edges in `Fearless Fly` reduces escape by Δ`motor.escape` $= -1.00$ ($202.4 \rightarrow 0.0$ Hz), while leaving chemotaxis steering ($-0.77/-0.71/-0.47$), forward walking (1.3 Hz), grooming (0.0 Hz), and feeding (19.0 Hz) **100.0% identical** to Wild-Type.
- Likewise, `Hyper-Escape`, `Obsessive Groomer`, and `Voracious Feeder` each reprogram their target sensorimotor reflex with **0.00 off-diagonal drift** across all other behavioral assays.

This zero-collateral-damage property is a direct consequence of two architectural features of evolved biological connectomes: (1) extreme graph sparsity ($\rho = 7.85 \times 10^{-4}$), and (2) anatomical bottlenecking into dedicated descending channels (`DNa01`/`DNa02`, `DNg97`, `DNp01`, `MDN`, `DNg62`, `MN9`) whose upstream 2-to-3 hop premotor relays are physically segregated across distinct neuropils.

2.6 Interactive 3D Connectome Surgery and Closed-Loop Embodied Platform

To make whole-brain connectome weight programming accessible for interactive exploration and hypothesis generation, we engineered an open-source, browser-based 3D laboratory (`https://statsleelab.github.io/connectome-weight-edit/`). The platform executes the complete 138,639-neuron FlyWire v783 CSR spiking engine inside a dedicated Web Worker coupled at 15 ms intervals to the 48-actuator NeuroMechFly 2.1 MuJoCo WebAssembly physics engine. Users can apply any of the six presets or perform custom 1–3 hop surgical pathway edits between arbitrary sensory and descending pools—independently tuning excitatory potentiation ($\gamma_{\text{exc}} \in [0, 8\times]$), inhibitory scaling ($\gamma_{\text{inh}} \in [0, 3\times]$), and direct synapse abliteration—while visualizing the exact modified synaptic pathways rendered in 3D anatomical brain space (potentiated synapses highlighted in cyan `#4ef2e0` and suppressed/ablated synapses in magenta `#ff5e7e`).

3. Discussion

By uniting concepts from LLM mechanistic interpretability with whole-brain computational neurobiology, this study demonstrates that a 138,639-neuron biological connectome can be interrogated, debugged, and surgically reprogrammed directly in weight space.

Bridging the EM Structure–Function Gap via Algorithmic Weight Calibration. Since the landmark C. elegans connectome and the recent adult *Drosophila* FlyWire reconstruction [2, 8, 10], a recurring critique has been that static anatomical synapse counts cannot explain dynamic behavior without physiological parameters [14]. Our mechanistic diagnosis of the Wild-Type left-turning bias provides a concrete quantitative resolution to this debate. We showed that the failure of raw EM simulations to perform bilateral chemotaxis is not a failure of the LIF spiking formalism, nor a missing pathway in the FlyWire reconstruction: the exact 3-hop anatomical bridge (21 left PNs $\rightarrow$ 61 left relays $\rightarrow$ `turn_left`; 56 right PNs $\rightarrow$ 61 right relays $\rightarrow$ `turn_right`) is physically present in the EM graph. Rather, the behavioral failure stems from equating raw synapse count N_{syn} with functional synaptic efficacy W_{uv}, which ignores both the physiological release-probability asymmetry discovered by Gaudry et al. [15] and the state-dependent neuromodulatory relief of tonic inhibitory clamps (`#112815` $\rightarrow$ `#904` and `#15988`/`#62170`/`#53459` $\rightarrow$ `DNg97`). Weight-space back-tracing provides a systematic algorithm to identify and calibrate these exact physiological degrees of freedom.

In Silico Predictions for Experimental Neurogenetics. Because every neuron in our CSR tensor maps one-to-one to a FlyWire v783 coordinate and cell identity, our weight-editing discoveries generate falsifiable predictions for wet-lab *Drosophila* neurogenetics:

1. Silencing or optogenetically hyperpolarizing the contralateral tonic inhibitory neuron `#112815` (`side` $= 1 \rightarrow$ `DNa02_R` `#904`, $W = -216$) should selectively disinhibit right-ward odor steering without impairing escape or grooming.

2. Selective genetic tetanus toxin (TNT) blockade of the 198 inhibitory inputs onto `DNg62` (`#26421`, `#65709`) should convert sub-threshold antennal mechanosensory noise into constitutive grooming behavior.

3. The asymmetric PN bottleneck (21 left vs. 56 right bridge PNs) predicts that homeostatic synaptic scaling at the LAL premotor relay (`Relay` $\rightarrow$ `DNa01/02`) is essential in vivo to compensate for developmental variability in glomerulus-to-LH projection counts.

Architectural Lessons for Modular NeuroAI. Finally, our side-by-side comparison between dense Transformer weight editing and sparse connectome surgery highlights a critical design principle for future artificial intelligence systems [1]. Whereas dense multilayer perceptrons entangle capabilities in polysemantic superposition—making surgical post-deployment editing fragile—the biological connectome routes information through sparse, anatomically segregated multi-hop bottlenecks equipped with dedicated inhibitory gates. Incorporating connectome-inspired sparse routing and explicit inhibitory clamp motifs into foundation model architectures could enable AI systems whose capabilities can be audited, calibrated, or abliterated with the same 0.0% collateral interference demonstrated here in the fly brain.

4. Methods

4.1 Whole-Brain LIF Spiking Formulation on CSR Tensors

Let $\mathcal{V} = \{0, 1, \ldots, N-1\}$ ($N = 138,639$) denote the set of neurons and $\mathcal{E} = \{0, 1, \ldots, M-1\}$ ($M = 15,091,983$) denote the directed edges stored in CSR arrays $\texttt{offsets} \in \mathbb{U}_{32}^{N+1}$, $\texttt{targets} \in \mathbb{U}_{32}^{M}$, and $\texttt{weights} \in \mathbb{I}_{16}^{M}$. Prior to any edit, we snapshot the pristine wild-type weight vector $\mathbf{W}^{(0)} \in \mathbb{I}_{16}^{M}$ so that all surgical interventions are computed deterministically relative to $\mathbf{W}^{(0)}$.

At each discrete integration step $t_k = k\Delta t$ ($\Delta t = 0.1$ ms), the membrane potential $V_i(t_k)$ and synaptic current $I_i(t_k)$ of neuron $i \in \mathcal{V}$ outside its refractory window ($t_k \geq t_i^{\text{ref}}$) evolve via exponential Euler integration:

$$V_i(t_{k+1}) = V_{\text{rest}} + (V_i(t_k) - V_{\text{rest}})e^{-\Delta t/\tau_{\text{mem}}} + \frac{\Delta t}{\tau_{\text{mem}}} I_i(t_k) \tag{5}$$

$$I_i(t_{k+1}) = I_i(t_k)e^{-\Delta t/\tau_{\text{syn}}} \tag{6}$$

When $V_i(t_{k+1}) \geq V_{\text{th}} = -45.0$ mV, neuron i emits a spike, resets to $V_i \leftarrow V_{\text{reset}} = -52.0$ mV, enters refractoriness until $t_i^{\text{ref}} = t_{k+1} + \tau_{\text{refrac}}$ ($\tau_{\text{refrac}} = 2.2$ ms), and schedules delayed postsynaptic current arrival at step $k_{\text{arr}} = k + 1 + \lfloor d_{\text{ax}}/\Delta t \rfloor$ (18 steps = 1.8 ms). Upon arrival at step k_{arr}, for every outgoing CSR edge $e \in [\texttt{offsets}[i], \texttt{offsets}[i+1])$ with target $j = \texttt{targets}[e]$, the postsynaptic current is updated by:

$$I_j(t_{k_{\text{arr}}}) \leftarrow I_j(t_{k_{\text{arr}}}) + \alpha_{\text{syn}} W_e, \qquad \alpha_{\text{syn}} = 0.275 \text{ mV} \tag{7}$$

Sensory inputs to pool $\mathcal{S}_m \subset \mathcal{V}$ with firing rate f_m (Hz) are injected at each step Δt as independent Bernoulli trials with probability $p_m = f_m \Delta t \times 10^{-3}$, adding $I_{\text{poisson}} = 250.0 \times \alpha_{\text{syn}} = 68.75$ mV per sensory event.

4.2 Spike-Weighted Synaptic Flux Attribution

To diagnose functional bottlenecks in the recurrent spiking network, we record the cumulative spike count vector $\mathbf{s}(\Delta T) \in \mathbb{N}_0^N$ over an evaluation horizon $\Delta T = K\Delta t$ ($K = 1,800$ steps = 180 ms). For any target neuron or readout pool $\mathcal{D} \subset \mathcal{V}$, we construct the transpose CSR incoming adjacency list $\text{InEdges}(\mathcal{D}) = \{e = (u \to v) \in \mathcal{E} : v \in \mathcal{D}\}$ and compute the per-edge spike-weighted synaptic flux:

$$\Phi_e(\Delta T) = s_u(\Delta T) \cdot W_e^{(0)} \tag{8}$$

Decomposing $\Phi_{\text{net}}(\mathcal{D})$ into excitatory ($\Phi_e > 0$) and inhibitory ($\Phi_e < 0$) components separates active excitatory drive from active inhibitory clamping. Sorting $\text{InEdges}^{-}(\mathcal{D})$ in ascending order of Φ_e isolates tonic inhibitory clamp neurons (such as $u = \texttt{\#112815} \to v = \texttt{\#904}$ with $\Phi_{12284124} = -4,536$) that remain invisible to static degree or weight histograms.

4.3 Bidirectional Multi-Hop Circuit Discovery and Surgical Weight Operators

Given a source sensory pool $\mathcal{S} \subset \mathcal{V}$, a target readout pool $\mathcal{D} \subset \mathcal{V}$, and a hop budget $H \in \{1, 2, 3\}$, our bidirectional pathway discovery algorithm (`ConnectomeEditor.discoverPathway`) computes the union of forward-reachable vertices within H hops of $\mathcal{S}$, $\mathcal{F}_h = \text{Succ}^h(\mathcal{S})$ ($0 \leq h \leq H$), and

backward-reachable vertices within H hops of $\mathcal{D}$ via the transpose CSR graph, $\mathcal{B}_{H-h} = \mathrm{Pred}^{H-h}(\mathcal{D})$. An edge $e = (u \to v) \in \mathcal{E}$ belongs to the H-hop causal bridge $\mathcal{E}_{\mathrm{bridge}}(\mathcal{S}, \mathcal{D}, H)$ if and only if for some $h \in \{0, \ldots, H-1\}$:

$$u \in \mathcal{F}_h \cap \mathcal{B}_{H-h} \quad \wedge \quad v \in \mathcal{F}_{h+1} \cap \mathcal{B}_{H-h-1} \tag{9}$$

For lateralized chemotaxis surgery (`Super-Forager`), we additionally enforce hemisphere concordance $\texttt{sides}[u] = \texttt{sides}[v] = h_{\mathrm{side}} \in \{1, 2\}$ on intermediate PN ($\mathcal{P}_h$) and LAL/LH relay ($\mathcal{R}_h$) vertices.

For each edge $e \in \mathcal{E}_{\mathrm{bridge}}$ with baseline weight $W_e^{(0)}$, the surgical weight transformation operator $\mathcal{T}$ updates the CSR weight tensor in place while preserving Int16 bounds $[-32767, +32767]$:

$$W_e \leftarrow \begin{cases} 0, & \text{if } \mathrm{Ablate}(e) \\ \mathrm{clip}(\lfloor \gamma_{\mathrm{exc}} W_e^{(0)} \rceil, 1, 32767), & \text{if } W_e^{(0)} > 0 \\ \mathrm{clip}(\lfloor \gamma_{\mathrm{inh}} W_e^{(0)} \rceil, -32767, -1), & \text{if } W_e^{(0)} < 0, \neg\mathrm{Dis}(e) \\ 0, & \text{if } W_e^{(0)} < 0, \mathrm{Dis}(e) \end{cases} \tag{10}$$

4.4 Embodied MuJoCo Biomechanical Coupling

Population firing rates r_k (Hz) of the seven descending/motor pools are converted into normalized motor commands $\mathbf{m} \in \mathbb{R}^6$:

$$m_{\mathrm{turn}} = \mathrm{clip}\left(\frac{r_{\mathrm{R}} - r_{\mathrm{L}}}{35} + \frac{0.45(r_{\mathrm{R}} - r_{\mathrm{L}})}{r_{\mathrm{R}} + r_{\mathrm{L}} + 8},\ -1, 1 \right) \tag{11}$$

$$m_{\mathrm{fwd}} = \mathrm{clip}\left(\frac{r_{\mathrm{fwd}}}{35} + \frac{r_{\mathrm{L}} + r_{\mathrm{R}}}{110},\ 0, 1 \right) \tag{12}$$

with analogous saturating transforms for $m_{\mathrm{esc}} = \min(1, r_{\mathrm{esc}}/45\ \mathrm{Hz})$, $m_{\mathrm{rev}} = \min(1, r_{\mathrm{rev}}/40\ \mathrm{Hz})$, $m_{\mathrm{groom}} = \min(1, r_{\mathrm{groom}}/38\ \mathrm{Hz})$, and $m_{\mathrm{feed}} = \min(1, r_{\mathrm{feed}}/35\ \mathrm{Hz})$. These commands modulate the amplitude, left–right stride differential, phase sign, and foreleg/proboscis posture of a 12 Hz tripod central pattern generator driving the 48 position-controlled actuators of the NeuroMechFly 2.1 model [13] in MuJoCo.

Data and Code Availability

All source code, CSR connectome weight-editing engines, quantitative benchmark suites, unit tests, and the interactive 3D WebGL/MuJoCo platform are openly available at `https://github.com/statsleelab/connectome-weight-edit`, with the live interactive laboratory hosted at `https://statsleelab.github.io/connectome-weight-edit/`. Companion LLM weight-space programming experiments are available at `https://github.com/statsleelab/weight_space_programming`.

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